

Problem 1) $y(x) = x$; $y(\alpha, x) = x + \alpha \sin x$; $\mathcal{F} = \left(\frac{dy}{dx}\right)^2$

$$\eta(x) = \sin x; \Rightarrow \frac{d\mathcal{F}(\alpha, x)}{d\alpha} = 1 + \alpha \sin x$$

$$\Rightarrow \mathcal{F} = 1 + 2\alpha \cos x + \alpha^2 \sin^2 x \Rightarrow \mathcal{F}(\alpha) = \int_0^{2\pi} (1 + 2\alpha \cos x + \alpha^2 \sin^2 x) dx$$

$$= 2\pi + \alpha^2 \pi \Rightarrow \boxed{\alpha = 0 \text{ at minimum.}}$$

Problem 2) $\mathcal{F}(\alpha) = \int_{x_1}^{x_2} \mathcal{F}(y(x), y'(x); x) dx$

$$\Rightarrow \frac{\partial \mathcal{F}}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial \mathcal{F}}{\partial y} \frac{\partial y}{\partial \alpha} + \frac{\partial \mathcal{F}}{\partial y'} \frac{\partial y'}{\partial \alpha} \right) dx; \quad \frac{\partial y}{\partial \alpha} = \eta(x); \quad \frac{\partial y'}{\partial \alpha} = \frac{d\eta}{dx}$$

$$\Rightarrow \frac{\partial \mathcal{F}}{\partial \alpha} = \int_{x_1}^{x_2} \left(\frac{\partial \mathcal{F}}{\partial y} \eta(x) + \frac{\partial \mathcal{F}}{\partial y'} \frac{d\eta(x)}{dx} \right) dx$$

Integration by parts gives us, $\int u dv = uv - \int v du$

$$\Rightarrow \int_{x_1}^{x_2} \frac{\partial \mathcal{F}}{\partial y'} \frac{d\eta}{dx} dx = \underbrace{\frac{\partial \mathcal{F}}{\partial y'} \eta(x)}_{=0 \text{ because } \eta(x_1) = \eta(x_2) = 0} - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial \mathcal{F}}{\partial y'} \right) \eta(x) dx$$

$$\Rightarrow \frac{\partial \mathcal{F}}{\partial \alpha} = \int_{x_1}^{x_2} \left[\frac{\partial \mathcal{F}}{\partial y} \eta(x) - \frac{d}{dx} \left(\frac{\partial \mathcal{F}}{\partial y'} \right) \eta(x) \right] dx$$

$$= \int_{x_1}^{x_2} \left[\frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \left(\frac{\partial \mathcal{F}}{\partial y'} \right) \right] \eta(x) dx$$

$$\frac{\partial \mathcal{F}}{\partial \alpha} = 0; \eta(x) \text{ is arbitrary} \Rightarrow \boxed{\frac{\partial \mathcal{F}}{\partial y} = \frac{d}{dx} \left(\frac{\partial \mathcal{F}}{\partial y'} \right)} \text{ Euler's eqn.}$$

Problem 3) Without loss of generality assume the coordinate origin is (x_1, y_1) . For simplicity, also assume the conservative force is downward along the x -axis, and the potential energy $U(x, y) = 0$. The kinetic energy is $T = \frac{1}{2} m v^2$; and the potential energy is $U = -mgx$ where g is the acceleration due to the conservative force. Thus, give $T + U = 0 \Rightarrow v = \sqrt{2gx}$

The time required for the transit is,

$$t = \int_{(x_1, y_1)}^{(x_2, y_2)} \frac{ds}{v} = \int_{x_1}^{x_2} \frac{(dx^2 + dy^2)^{1/2}}{(2gx)^{1/2}} = \int_{x_1}^{x_2} \left(\frac{1+y'^2}{2gx} \right)^{1/2} dx$$

We want t to be minimized. $\Rightarrow \mathcal{F} = \left(\frac{1+y'^2}{x} \right)^{1/2}$ (excluding g which is constant). The necessary condition from Euler's eqn is,

$$\frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} = 0 \Rightarrow \frac{\partial \mathcal{F}}{\partial y'} = \text{const.} = \frac{1}{\sqrt{2a}}$$

$$\Rightarrow \frac{1}{x} \left(\frac{1+y'^2}{x} \right)^{-1/2} (x'_x y') = \frac{1}{\sqrt{2a}}$$

$$\Rightarrow \frac{y'}{\sqrt{x} \sqrt{1+y'^2}} = \frac{1}{\sqrt{2a}} \Rightarrow \frac{y'^2}{x(1+y'^2)} = \frac{1}{2a}$$

$$y'^2 = \frac{1}{2a} (x + xy'^2) \Rightarrow \left(\frac{x}{2a} - 1 \right) y'^2 + \frac{x}{2a} = 0$$

$$y' = \sqrt{-\frac{x}{2a} \times \frac{2a}{x-2a}} = \sqrt{\frac{x}{2a-x}}$$

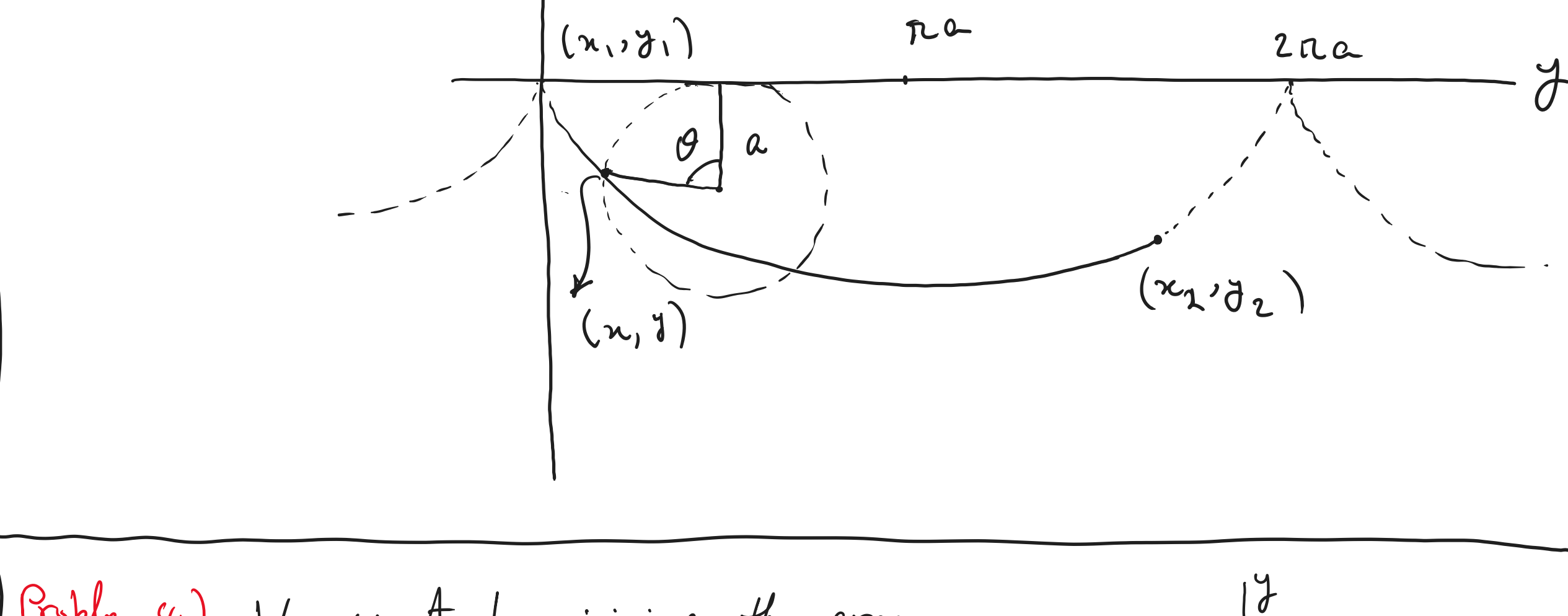
$$\int dy = \int \sqrt{\frac{x}{2a-x}} dx = \int \frac{x}{\sqrt{2ax-x^2}} dx$$

Assume, $x = a(1 - \cos \theta) \Rightarrow dx = a \sin \theta d\theta$

$$\Rightarrow y = \int a(1 - \cos \theta) d\theta \Rightarrow y = a(\theta - \sin \theta) + \text{const.}$$

$$\Rightarrow \begin{cases} x = a(1 - \cos \theta) \\ y = a(\theta - \sin \theta) \end{cases} \text{ This is the cycloid equation.}$$

Here, a must be set such that the path passes through the start and the end points.



Problem 4) We want to minimize the area.

$$dA = 2\pi x ds = 2\pi x (dx^2 + dy^2)^{1/2}$$

$$A = 2\pi \int_{x_1}^{x_2} x(1+y'^2)^{1/2} dx; \quad y' = \frac{dy}{dx}$$

$$\Rightarrow \begin{cases} \mathcal{F} = x(1+y'^2)^{1/2} \\ \frac{\partial \mathcal{F}}{\partial y} = 0; \quad \frac{\partial \mathcal{F}}{\partial y'} = \frac{xy'}{(1+y'^2)^{1/2}} \end{cases}$$

From Euler's eqn:

$$\frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} = 0$$

$$\Rightarrow \frac{d}{dx} \left[\frac{xy'}{(1+y'^2)^{1/2}} \right] = 0 \Rightarrow \frac{xy'}{(1+y'^2)^{1/2}} = \text{const.} \equiv a$$

$$\Rightarrow y' = \frac{a}{(x^2 - a^2)^{1/2}} \Rightarrow y = \int \frac{a}{(x^2 - a^2)^{1/2}} dx$$

$$\Rightarrow y = a \cosh^{-1} \left(\frac{x}{a} \right) + b \text{ where } a \text{ \& } b \text{ constants are determined by requiring the surface to pass through } (x_1, y_1), (x_2, y_2)$$

We can also write $x = a \cosh \left(\frac{y-b}{a} \right)$ which is the eqn of a catenary (i.e., the curve of a flexible cord hanging freely between two points)

Problem 5) $\frac{d\mathcal{F}}{dx} = \frac{\partial \mathcal{F}}{\partial y} \frac{dy}{dx} + \frac{\partial \mathcal{F}}{\partial y'} \frac{dy'}{dx} + \frac{\partial \mathcal{F}}{\partial x}$

$$= y' \frac{\partial \mathcal{F}}{\partial y} + y' \frac{\partial \mathcal{F}}{\partial y'} + \frac{\partial \mathcal{F}}{\partial x} \quad (1)$$

Also, note that, $\frac{d}{dx} \left(y' \frac{\partial \mathcal{F}}{\partial y'} \right) = y'' \frac{\partial \mathcal{F}}{\partial y'} + y' \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} \quad (2) \text{ chain rule}$

$$(1), (2) \Rightarrow \frac{d\mathcal{F}}{dx} - \frac{\partial \mathcal{F}}{\partial x} - y' \frac{\partial \mathcal{F}}{\partial y} + y' \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} = \frac{d}{dx} \left(y' \frac{\partial \mathcal{F}}{\partial y'} \right)$$

But, the Euler eqn is,

$$0 = \frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'}$$

while, $S = y' \left(\frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} \right) \Rightarrow S = 0$

$$\Rightarrow \boxed{\frac{\partial \mathcal{F}}{\partial x} - \frac{d}{dx} \left(\mathcal{F} - y' \frac{\partial \mathcal{F}}{\partial y'} \right) = 0} \text{ The second form of the Euler's eqn.}$$

Problem 6) The element of length on the surface of a sphere is,*

$$ds = r (d\theta^2 + \sin^2 \theta d\phi^2)^{1/2} \text{ where } r \text{ is the sphere radius.}$$

The distance between two points is, therefore,

$$s = r \int_1^2 \left(\left(\frac{d\theta}{d\phi} \right)^2 + \sin^2 \theta \right)^{1/2} d\phi$$

$$\Rightarrow \mathcal{F} = (\theta'^2 + \sin^2 \theta)^{1/2}; \quad \theta' = \frac{d\theta}{d\phi} \Rightarrow \frac{\partial \mathcal{F}}{\partial \phi} = 0$$

Therefore, we can use the second form of Euler's eqn.

$$\Rightarrow (\theta'^2 + \sin^2 \theta)^{1/2} - \theta' \frac{\partial}{\partial \theta'} (\theta'^2 + \sin^2 \theta)^{1/2} = \text{const.} \equiv a$$

$$\Rightarrow \sin^2 \theta = a (\theta'^2 + \sin^2 \theta)^{1/2} \Rightarrow \theta' = \frac{(1 - a \csc^2 \theta)^{1/2}}{a \csc^2 \theta}$$

$$\Rightarrow \frac{d\phi}{d\theta} = \frac{a \csc^2 \theta}{(1 - a \csc^2 \theta)^{1/2}}; \quad \csc \theta = \frac{1}{\sin \theta}$$

$$\Rightarrow \phi = \sin^{-1} \left(\frac{\cot \theta}{\beta} \right) + \alpha \text{ integration constant.}$$

$$\Rightarrow \cot \theta = \beta \sin(\phi - \alpha) \quad (3) \text{ This is in spherical coordinate.}$$

Recall $\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}; \quad \sin(\phi - \alpha) = \sin \phi \cos \alpha - \cos \phi \sin \alpha$

Let's multiply (3) by the factor $\rho \sin \theta$ and expand $\sin(\phi - \alpha)$ together,

$$r \cos \theta = \underbrace{(\rho \cos \alpha) r \sin \theta \sin \phi}_A - \underbrace{(\beta \sin \alpha) r \sin \theta \cos \phi}_B$$

$$\Rightarrow A(r \sin \theta \sin \phi) - B(r \sin \theta \cos \phi) = r \cos \theta$$

$$\Rightarrow Ay - Bx = z \text{ This is the eqn of a plane passing through the sphere center, which intersects the sphere to form circles. Thus, the shortest path on spheres is part of a circle.}$$

* Recall that the line element for an infinitesimal displacement from (r, θ, ϕ) to $(r+dr, \theta+d\theta, \phi+d\phi)$ is $d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$

Problem 7) $y = R\theta$

$$\Rightarrow \begin{cases} \mathcal{F}(y, \theta) \equiv y - R\theta = 0 \\ \frac{\partial \mathcal{F}}{\partial y} = 1; \quad \frac{\partial \mathcal{F}}{\partial \theta} = -R \end{cases}$$

Problem 8) We want to maximize the area under the curve.

$$\mathcal{J} = \int_{-a}^a y dx \text{ with constraints}$$

$$y(x): y(-a) = 0; y(a) = 0 \text{ AND}$$

$$K = \int dl = l$$

The differential length along the curve $dl = (dx^2 + dy^2)^{1/2} = (1+y'^2)^{1/2} dx$ where $y' = \frac{dy}{dx}$. The functional for the constraint be consed,

$$K = \int_{-a}^a (1+y'^2)^{1/2} dx = l; \quad y(x) = y; \quad g(x) = \sqrt{1+y'^2}$$

We want both $y(x)$ and $g(x)$ extremized.

Therefore, the combined (constrained) functional to extremize is,

$$\int_{-a}^a (\mathcal{F} + \lambda g) dx \text{ which requires the satisfaction of,}$$

$$\frac{\partial \mathcal{F}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{F}}{\partial y'} + \lambda \left(\frac{\partial g}{\partial y} - \frac{d}{dx} \frac{\partial g}{\partial y'} \right) = 0 \quad (1)$$

Here, $\frac{\partial \mathcal{F}}{\partial y} = 1, \quad \frac{\partial \mathcal{F}}{\partial y'} = 0, \quad \frac{\partial g}{\partial y} = 0, \quad \frac{\partial g}{\partial y'} = \frac{y'}{(1+y'^2)^{1/2}} \quad (2)$

$$(1), (2) \Rightarrow 1 - \lambda \frac{d}{dx} \left[\frac{y'}{(1+y'^2)^{1/2}} \right] = 0$$

$$\Rightarrow d \left[\frac{y'}{(1+y'^2)^{1/2}} \right] = dx \Rightarrow \frac{\lambda y'}{(1+y'^2)^{1/2}} = x - C_1$$

$$\Rightarrow dy = \frac{\pm (x - C_1) dx}{\sqrt{\lambda^2 - (x - C_1)^2}} \Rightarrow y = \mp \sqrt{\lambda^2 - (x - C_1)^2} + C_2$$

$$\Rightarrow (x - C_1)^2 + (y - C_2)^2 = \lambda^2 \text{ This is the equation of a circle which in this problem is bounded from the bottom by the condition } y=0.$$